Partial PWR hierarchy for merged ERC/netlist. NOT a full board, PCB or fabrication release.

pwr_mcu_interface

Dosya: pwr_mcu_interface.kicad_sch

k2_reference

Dosya: k2_reference.kicad_sch

actuator_path

Dosya: actuator_path.kicad_sch

safety_gate

Dosya: safety_gate.kicad_sch

arming_io

Dosya: arming_io.kicad_sch

INA228, regulators, TVS/regen clamp and PCB missing: not a full board

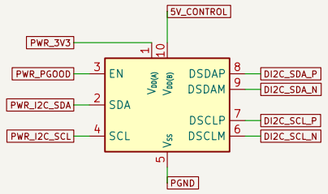
Sheet: /
File: ugv_pwr_01.kicad_sch

Title: UGV-PWR-01 partial board hierarchy

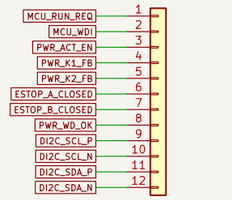
Size: A4	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 1/6

PWR board — local PCA9615

U501
PCA9615DP

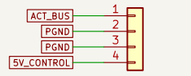


J501

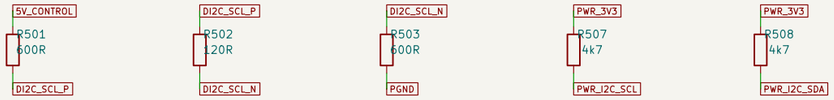


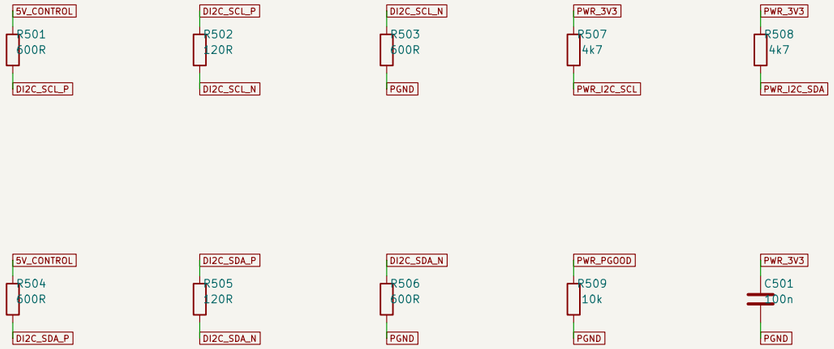
BM12B-GHS-TBT

J502



768250004





#FLG501
PWR_FLAG
PWR_3V3

#FLG502
PWR_FLAG
5V_CONTROL

#FLG503
PWR_FLAG
PGND

BOARD INTERFACE DRAFT: safety nets 1–8 and ACT_BUS continue on other sheets

J2 pins 2/3 join at PGND; routing/current sharing and mating CAD remain open

3V3 regulator and PGOOD source are external; not a release-to-fab schematic

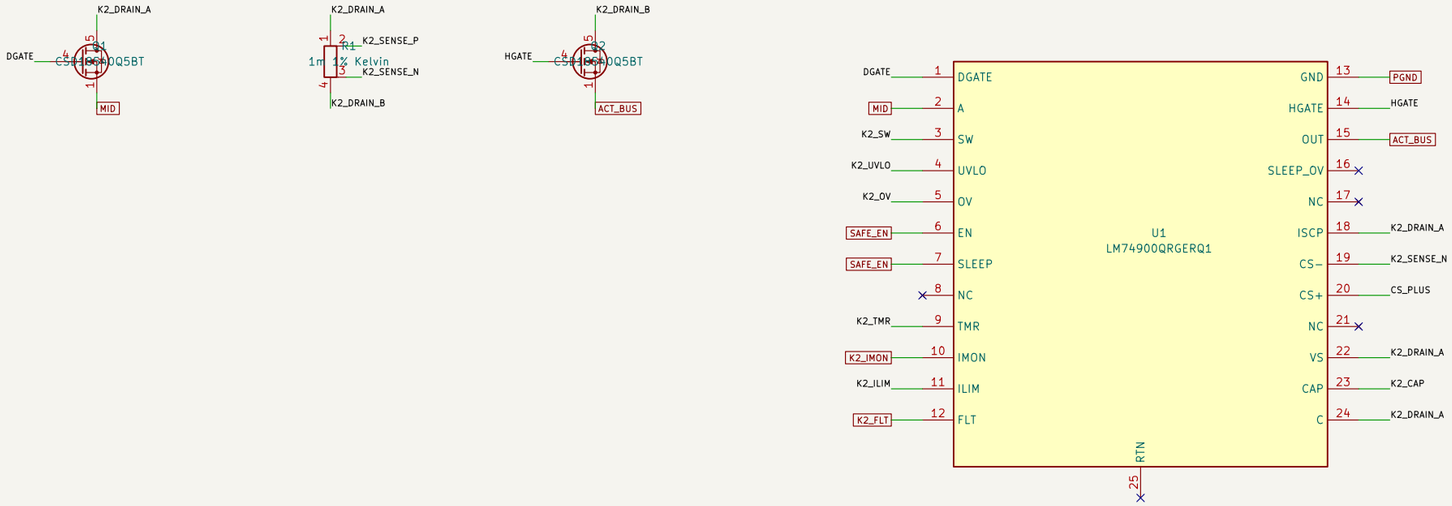
Partial interface sheet; not a production board schematic

Sheet: /pwr_mcu_interface/
File: pwr_mcu_interface.kicad_sch

Title: UGV-01 board interface draft

Size: A3	Date: 2026-09-23	**Rev: P1**
KiCad E.D.A. 10.0.6		Id: 2/6

K2 dual-FET separator — reference circuit, NOT for fabrication



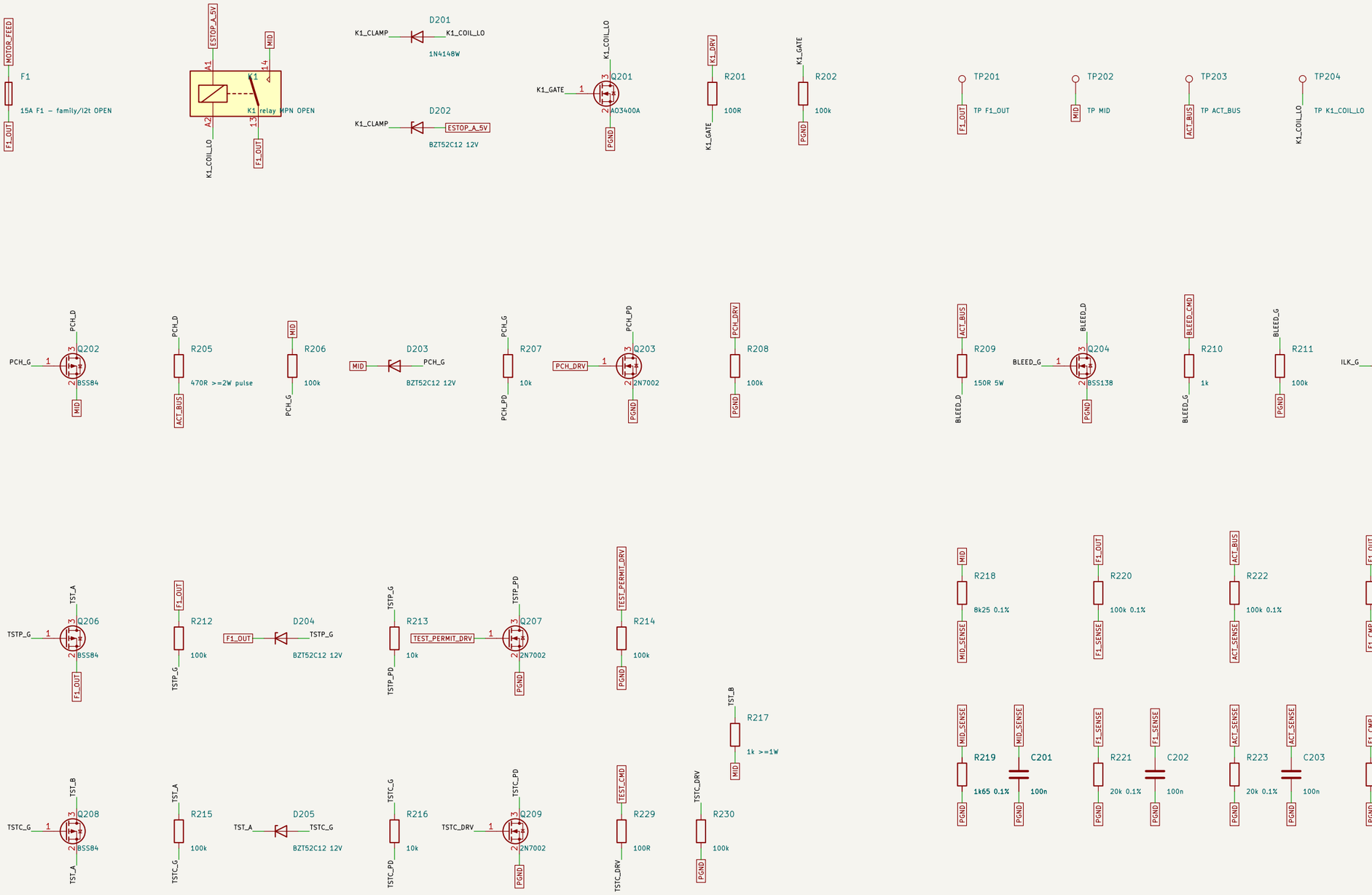
K1/precharge/bleed/test: actuator_path; SAFE_EN driver: safety_gate; K2_FLT/IMON: arming_io. Clamp/TVS, FLT latch input, shunt MPN OPEN

U1 pad 25: solder to isolated copper island; NO PGND connection (TI Rev C, Table 5-1)

MID -> Q1 -> R1 Kelvin shunt -> Q2 -> ACT_BUS; power nets are labels, not wire-width proof

Conditional circuit only; no PWR board or safety release

Sheet: /k2_reference/		
File: k2_reference.kicad_sch		
Title: UGV-PWR-01 K2 reference		
Size: A3	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 3/6



P0 REVIEW SHEET — NOT FOR FABRICATION. K1 MPN/footprint, F1 I2t, R205/R209/R217 parts, TVS and regen clamp OPEN.

K2 (LM74900 + 2x CSD18540, Kelvin shunt, 4700uF) is on k2_reference; MID -> K2 -> ACT_BUS.

K1 coil supply = ESTOP_A_5V (NC-A contact); Q201 low side needs SAFE_GATE AND K1_CMD (safety_gate).

Precharge/test FET gates are driven from ESTOP_B_5V via LVC1T45: NC-B open removes the drive physically.

Test path bypasses K1 only; K2/precharge short is detected before K1 closes (MCU sequence, not HW interlock).

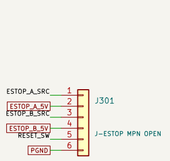
Small-signal FET ratings not verified against vendor datasheets in this revision.

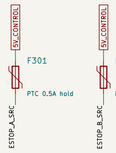
Actuator path P0: K1 selection open; not a PCB release

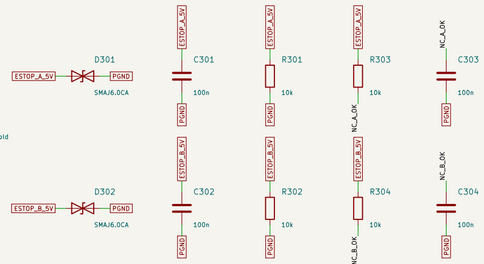
Sheet: /actuator_path/
File: actuator_path.kicad_sch

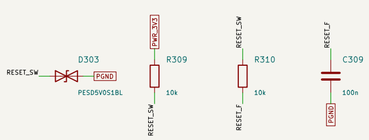
Title: UGV-PWR-01 actuator path: F1-K1-MID-K2-ACT_BUS

Size: A3	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 4/6

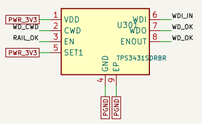


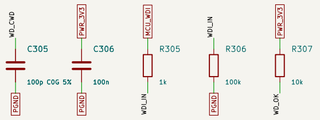


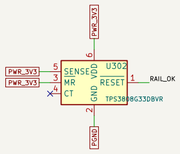


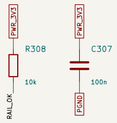


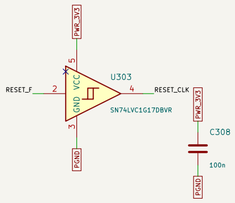


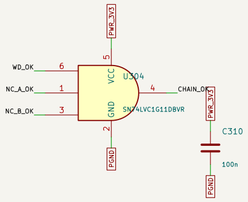


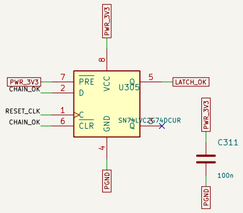


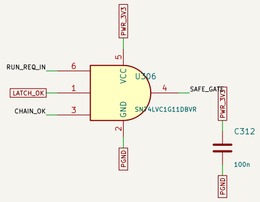


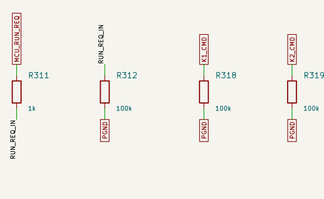


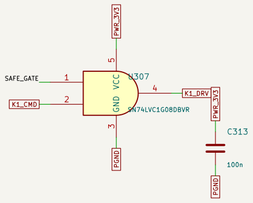


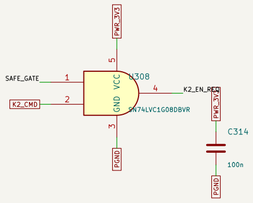


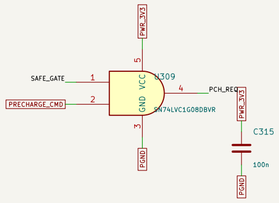


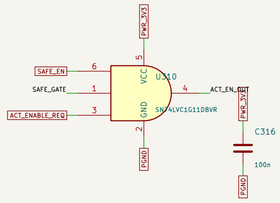


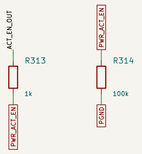


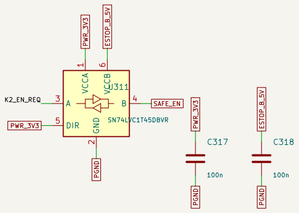


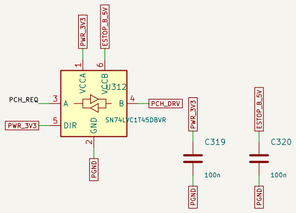


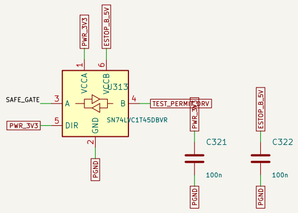


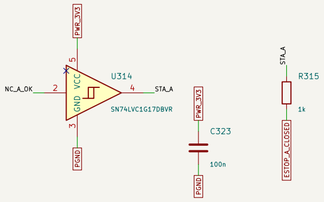


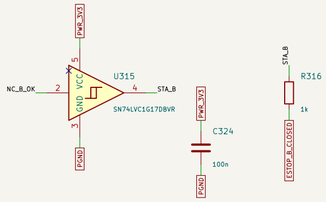


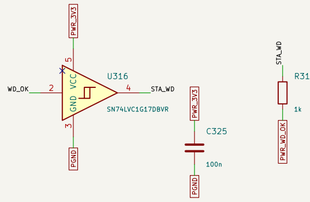


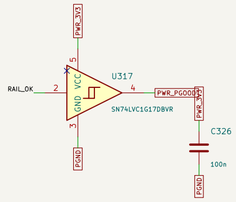












P0 REVIEW SHEET — NOT FOR FABRICATION: not a functional-safety certification.

CHAIN_OK = NC_A AND NC_B AND WD_OK; WD_OK = TPS3431 WDO AND ENOUT (wired); TPS3808 RAIL_OK drives TPS3431 EN.

SAFE_GATE = LATCH_OK AND CHAIN_OK AND MCU_RUN_REQ. Latch sets only on reset-button release with CHAIN_OK High.

NC-A carries K1 coil energy; NC-B powers the LVC1T45 B side (K2 EN, precharge, test permit). Logic cannot veto an open NC.

LM74900 SLEEP is not an E-stop: SLEEP=EN=SAFE_EN, driven Low by hardware. K2_FLT is read by MCU only (latch input OPEN).

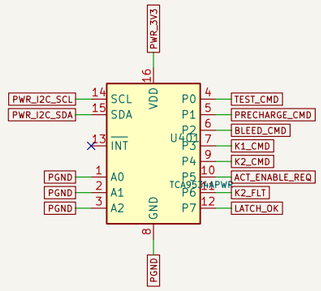
J-ESTOP, PTC, TVS MPNs OPEN. T01/T03/T13 physical evidence NOT_RUN.

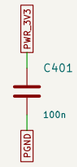
Safety permit P0: not a PCB or functional-safety release

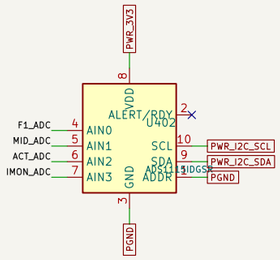
Sheet: /safety_gate/
File: safety_gate.kicad_sch

Title: UGV-PWR-01 dual NC E-stop, watchdog and permit chain

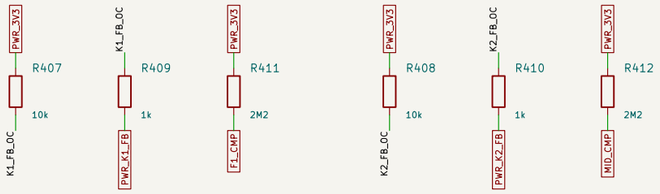
Size: A2	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 5/6

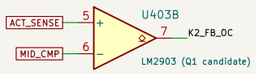


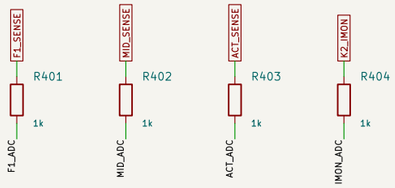


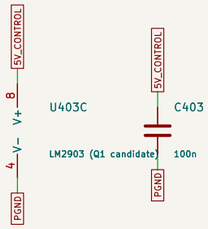












P0 REVIEW SHEET — NOT FOR FABRICATION. TCA9534A 0x38, ADS1115 0x48 on PWR_I2C behind PCA9615.

TCA outputs power up as inputs; consumer-side 100k pulldowns hold commands Low. Stuck-High commands are gated by SAFE_GATE.

ADS1115 is for ARMING diagnosis only — never the E-stop or OV cut path. AIN3 = K2 IMON.

LM2903 feedback reports measured state across K1 and K2; MCU compares with command and time.

Arming I/O P0: not a PCB release

Sheet: /arming_io/
File: arming_io.kicad_sch

Title: UGV-PWR-01 arming commands, rail ADC and K1/K2 feedback

Size: A3	Date: 2026-09-24	**Rev: P0**
KiCad E.D.A. 10.0.6		Id: 6/6